

Case Study: Agri-Food Data Analytics & BI

Dashboarding

Project Title:	National Agri-Food Supply Chain & Food Desert Accessibility Risk Engine
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Tools & Tech:	Google Looker Studio, Google Sheets / Relational Data Modeling, SQL Conditional Logic, Agribusiness Data Analytics
Data Source:	agri_food_supply_chain_data.csv (Multi-Province Agricultural Production & Regional Supply-Demand Database, 2021–2026)
District Clusters:	Punjab North (Faisalabad), Sindh Central (Sukkur), KPK Valley (Peshawar)
Primary Commodities:	Wheat (Punjab), Rice (Sindh), Maize (KPK)
Time Horizon:	2021 – 2026 Annual Crop Cycles

1. Project Background & Problem Statement:

- **The Real-World Agronomic Problem**

Agri-food supply chains in developing agricultural economies face severe structural imbalances due to climate-induced yield volatility, poor post-harvest distribution logistics, and socio-economic access barriers. Traditional agricultural monitoring systems often track crop production in isolation without linking localized crop deficits directly to geographic accessibility (distance to market hubs) and population vulnerability (poverty rates). By the time localized food shortages and price spikes hit rural and peri-urban markets, intervention options become limited and expensive.

- **The Solution:**

By integrating relational agricultural supply-demand datasets with multi-dimensional SQL risk logic inside Google Looker Studio, we engineered a dynamic business intelligence dashboard. This system quantifies net supply deficits, tracks multi-year crop yield swings against regional

demand, and automatically classifies regional clusters into Food Desert Risk Tiers to support targeted supply chain interventions and emergency grain logistics.

2. Data Source & Technical Architecture:

Dataset Features (agri_food_supply_chain_data.csv):

- **Spatial & Commodity Identifiers:** district_cluster, province, primary_crop.
- **Volumetric Metrics:** actual_yield_tons, estimated_demand_tons, supply_deficit_tons.
- **Socio-Climate Indicators:** drought_risk_index (**0.0 - 1.0**), dist_to_grocery_hub_km (km), low_income_pop_pct (%).
- **Engineered Risk Classification:** food_desert_risk_tier.

Software Stack:

- **Google Sheets / Data Architecture:** Relational schema design, dataset cleaning, aggregate calculations, and formula validation.
- **Google Looker Studio:** Cloud BI environment for interactive filtering, scorecards, time-series visual trend mapping, and heatmap risk matrices.

3. Technical Methodology & Workflow:

Step 1: Net Supply Deficit Calculation:

Derived the net structural supply gap per district cluster and crop year:

$$\text{Supply Deficit (Tons)} = \text{Estimated Demand (Tons)} - \text{Actual Yield (Tons)}$$

Across the complete multi-year baseline (18 valid records), total crop yield stands at **166,900 Tons** against an estimated consumer demand of **189,900 Tons**, yielding a cumulative net structural deficit of **23,000 Tons**.

Step 2: Multidimensional Risk Engine (SQL Logic)

Engineered a conditional classification algorithm to categorize regional food desert vulnerability based on net deficit (**>1,500 tons**), spatial isolation (**>20 km to distribution hubs**), and economic vulnerability (**>40 low-income population**):

CASE

WHEN supply_deficit_tons > 1500 AND dist_to_grocery_hub_km > 20 AND
low_income_pop_pct > 40

THEN 'Critical Food Desert Risk'

WHEN supply_deficit_tons > 0

THEN 'Moderate Supply Vulnerability'

ELSE 'Stable Access Zone'

END AS food_desert_risk_tier

4. Results & Dashboard Visual Analytics:



Figure 1: Interactive Google Looker Studio Agri-Food Risk Dashboard Overview.

A. Executive KPI Scorecards (Filtered View: Crop Year 2026):

The Looker Studio dashboard primary header presents aggregated metrics for the 2026 crop cycle:

KPI Scorecard Metric	Dashboard Display Value	Exact Mathematical Derivation	Agronomic & Operational Interpretation
supply_deficit_tons	3,300	$2,000 \text{ (Punjab)} + 800 \text{ (Sindh)} + 500 \text{ (KPK)}$	Total annual shortfall across all three commodity belts.
drought_risk_index	0.39	$\text{Average } (0.48 + 0.38 + 0.32) / 3 = 0.3933$	Favorable baseline climate condition across regional clusters.
dist_to_grocery_hub_km	15.83	$\text{Average } (11.0 + 20.0 + 16.5) / 3 = 15.833 \text{ km}$	Average last-mile logistics distance to distribution hubs.

(Note: Across the full unfiltered 6-year database, cumulative deficit is **23,000 Tons**, mean drought risk is **0.53**, and mean distance to grocery hub is **18.5 km**).

B. Multi-Year Crop Yield Volatility Trends (2021–2026)

The Looker Studio Time-Series Chart tracks actual annual crop yield trajectories (**Tons**) across regional clusters:

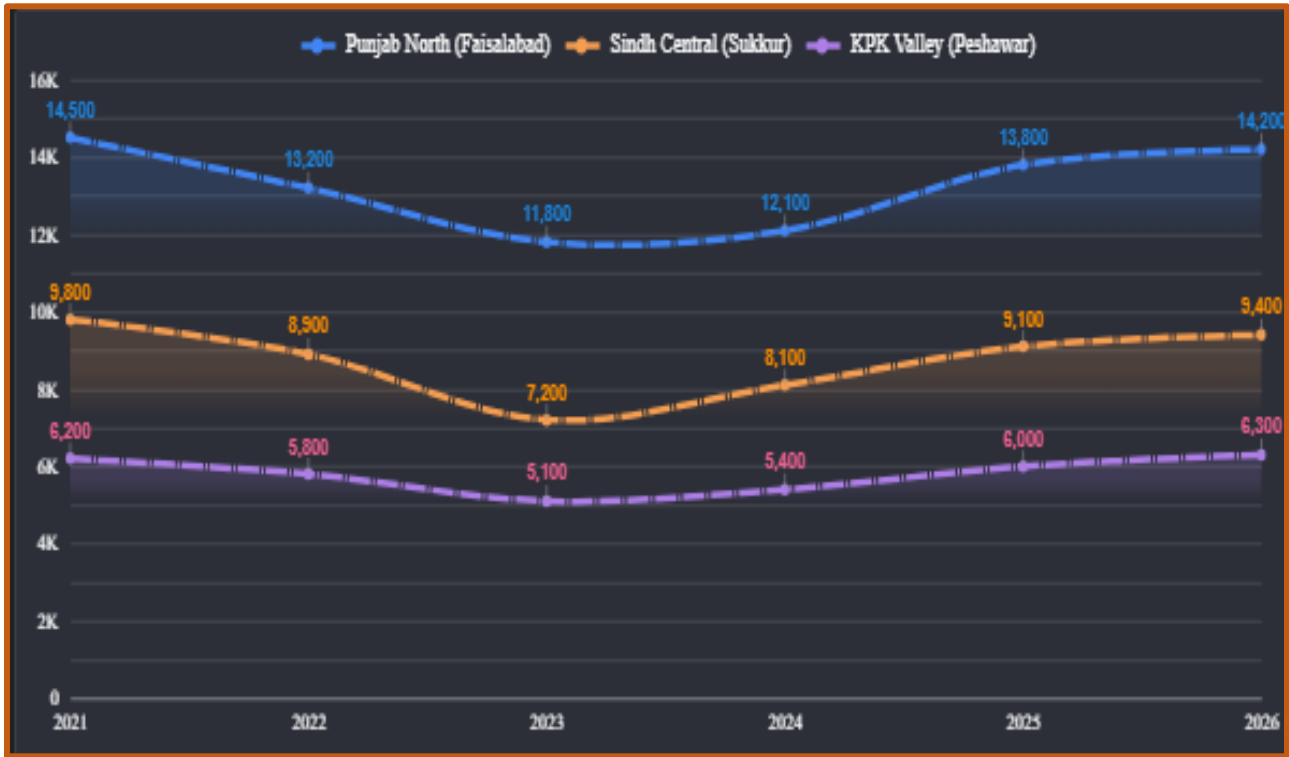


Figure 2: Multi-Year Crop Yield Trajectory across Regional Clusters (2021–2026)

Crop Year	Punjab North (Wheat)	Sindh Central (Rice)	KPK Valley (Maize)
2021	14,500 Tons	9,800 Tons	6,200 Tons
2022	13,200 Tons	8,900 Tons	5,800 Tons
2023 (Drought Peak)	11,800 Tons (Dip)	7,200 Tons (Dip)	5,100 Tons (Dip)
2024	12,100 Tons	8,100 Tons	5,400 Tons
2025	13,800 Tons	9,100 Tons	6,000 Tons
2026	14,200 Tons	9,400 Tons	6,300 Tons

Key Trend Insights:

- **The 2023 Climate Drop:** All three regional clusters hit their lowest yield levels in 2023 due to peak drought risk indices (**0.82** in Punjab, **0.88** in Sindh, **0.71** in KPK).
- **Punjab North (Faisalabad - Wheat):** Maintains high production volume (**11,800 - 14,500 tons**), but faces persistent urban demand growth (**15,000 - 16,200 tons**), leading to continuous deficit.
- **Sindh Central (Sukkur - Rice):** Showed severe vulnerability in 2023 (**7,200 tons yield vs 9,500 tons demand**) with distance to market expanding to **28.5 km** and poverty at **52%**, triggering **Critical Food Desert Risk** status.
- **KPK Valley (Peshawar - Maize):** Demonstrates lower overall volume (**5,100 - 6,300 tons**), operating under narrow margins with steady recovery toward 2026.

5. Practical Value & Agribusiness Applications:

- **Early Warning Logistics Allocation:** Flags structural deficit zones early, allowing provincial food departments to redirect grain reserves before market shortages occur.
- **Targeted Supply Chain Infrastructure:** Helps government planning agencies identify where to construct new wholesale storage centers and distribution hubs based on distance-to-market data (**>20km**).
- **Policy & Social Safety Nets:** Provides a multi-dimensional risk index for targeting food subsidy distributions to verified food desert areas.

6. Conclusion:

This project demonstrates how combining relational data engineering with dynamic cloud BI dashboarding in Google Looker Studio converts complex multi-source agricultural datasets into actionable supply chain intelligence. The automated risk classification matrix accurately isolates

food desert hotspots, providing a scalable decision-support platform for climate-smart agribusiness logistics and regional food security policy.

7. Project Deliverables & Interactive Links:

- **Live Interactive Dashboard:**

<https://datastudio.google.com/reporting/920b3d9a-05e3-4924-b0ad-33ba72ab967a/page/cuk5F>

- **Dataset Source:**

agri_food_supply_chain_data.csv